```
clc; clear; close all;

%% IIR Filter Coefficients
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

N = 50;
x = [1 zeros(1,N-1)]; % Unit impulse
n = 0:N-1;

%% 1. Direct Form-I and Direct Form-II
y1 = filter(b,a,x); % Direct Form-I
y2 = filter(b,a,x); % Direct Form-II

%% 2. Cascade Realization
[sos,g] = tf2sos(b,a);
yc = g*x;

for k = 1:size(sos,1)
    yc = filter(sos(k,1:3),sos(k,4:6),yc);
end

%% 3. Parallel Realization
[r,p,k] = residuez(b,a);
yp = filter(k,1,x);

for i = 1:length(r)
    yp = yp + filter(r(i),[1 -p(i)],x);
end

%% 4. Display Coefficients
disp('Numerator coefficients b ='); disp(b)
```

```
Numerator coefficients b =
    0.0048    0.0193    0.0289    0.0193    0.0048
```

```
disp('Denominator coefficients a ='); disp(a)
```

```
Denominator coefficients a =
    1.0000   -2.3695    2.3140   -1.0547    0.1874
```

```
disp('Second Order Sections (Cascade) ='); disp(sos)
```

```
Second Order Sections (Cascade) =
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327
```

```
disp('Cascade Gain ='); disp(g)
```

```
Cascade Gain =
    0.0048
```

```
disp('Residues (Parallel) ='); disp(r)
```

```
Residues (Parallel) =
  -0.2698 + 0.1343i
  -0.2698 - 0.1343i
   0.2594 - 0.8024i
   0.2594 + 0.8024i
```

```
disp('Poles (Parallel) ='); disp(p)
```

```
Poles (Parallel) =
   0.6604 + 0.4433i
   0.6604 - 0.4433i
   0.5243 + 0.1459i
   0.5243 - 0.1459i
```

```
disp('Direct term ='); disp(k)
```

```
Direct term =
    0.0256
```

```
%% 5. Error and Percentage Deviation
E = [max(abs(y1-y2));
    max(abs(y1-yc));
    max(abs(y1-yp))];

D = E/max(abs(y1))*100;

T = table(["DF-I";"DF-II";"Cascade";"Parallel"], ...
    [0;0;E(2);E(3)], ...
    [0;0;D(2);D(3)], ...
    'VariableNames', ...
    {'Structure','Maximum_Error','Deviation_percent'});

disp('Comparison Table:')
```

```
Comparison Table:
```

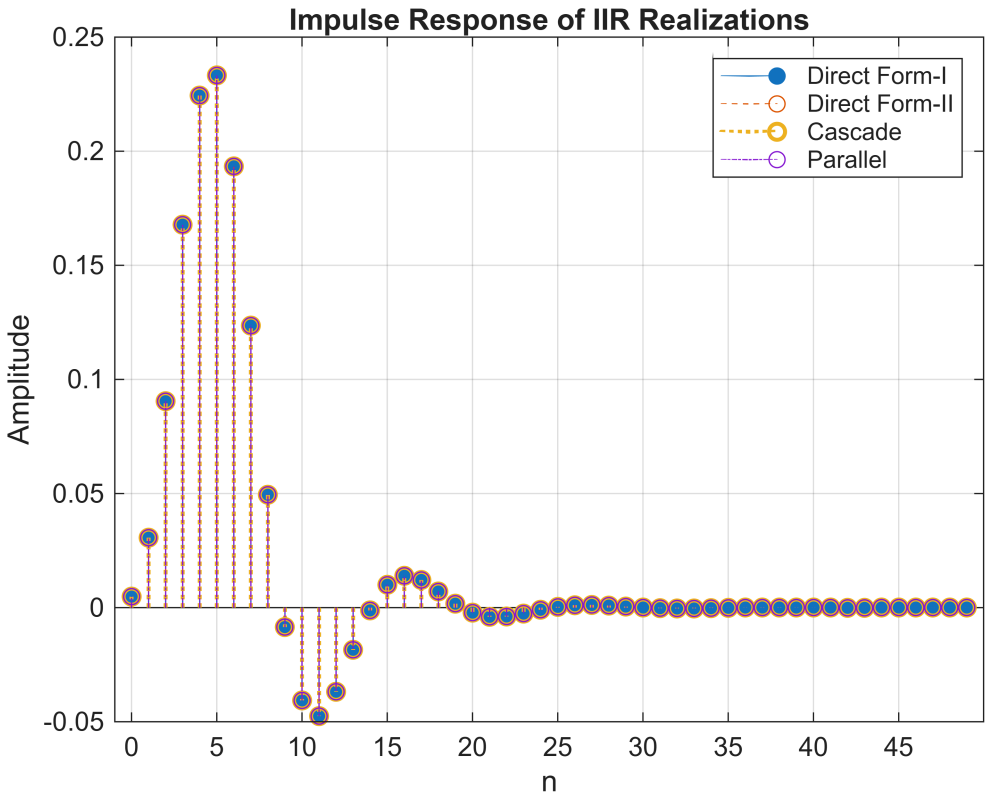
```
disp(T)
```

Structure	Maximum_Error	Deviation_percent
"DF-I"	0	0
"DF-II"	0	0
"Cascade"	1.9221e-15	8.242e-13
"Parallel"	1.9811e-15	8.4949e-13

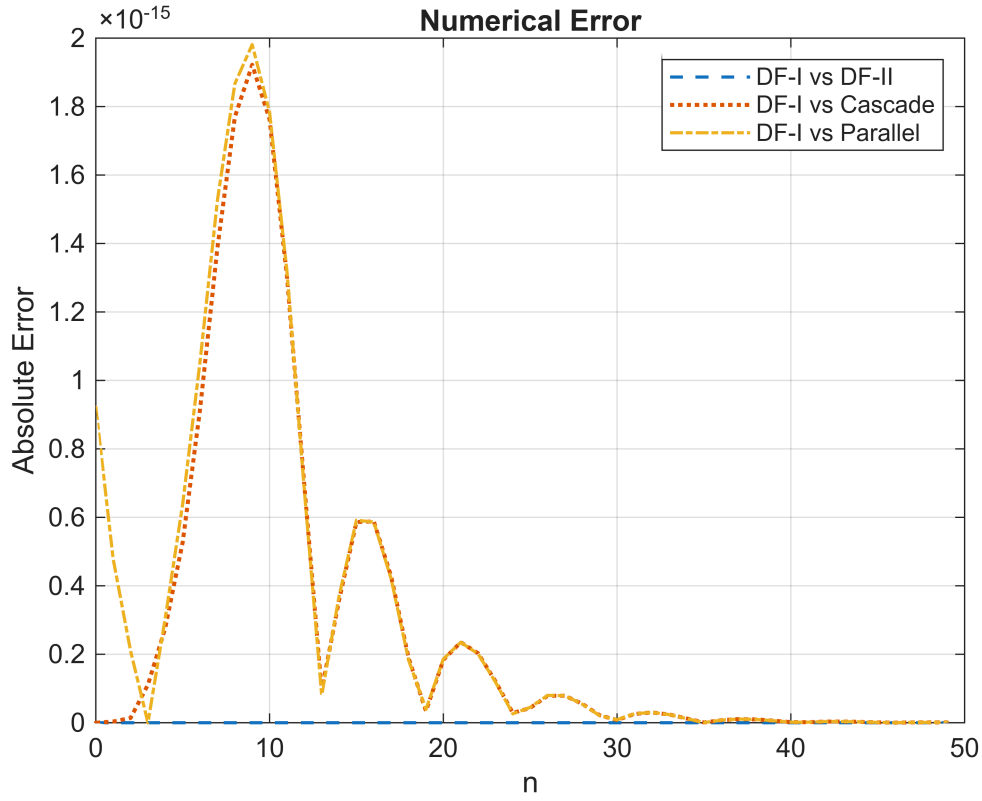
```
%% 6. Impulse Response Comparison
figure;
stem(n,y1,'filled'); hold on;
stem(n,y2,'--');
stem(n,yc,':','LineWidth',1.5);
stem(n,yp,'-.');
```

```
Warning: Using only the real component of complex data.
```

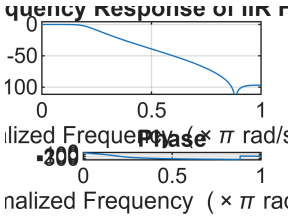
```
grid on;
xlabel('n');
ylabel('Amplitude');
title('Impulse Response of IIR Realizations');
legend('Direct Form-I','Direct Form-II','Cascade','Parallel');
```



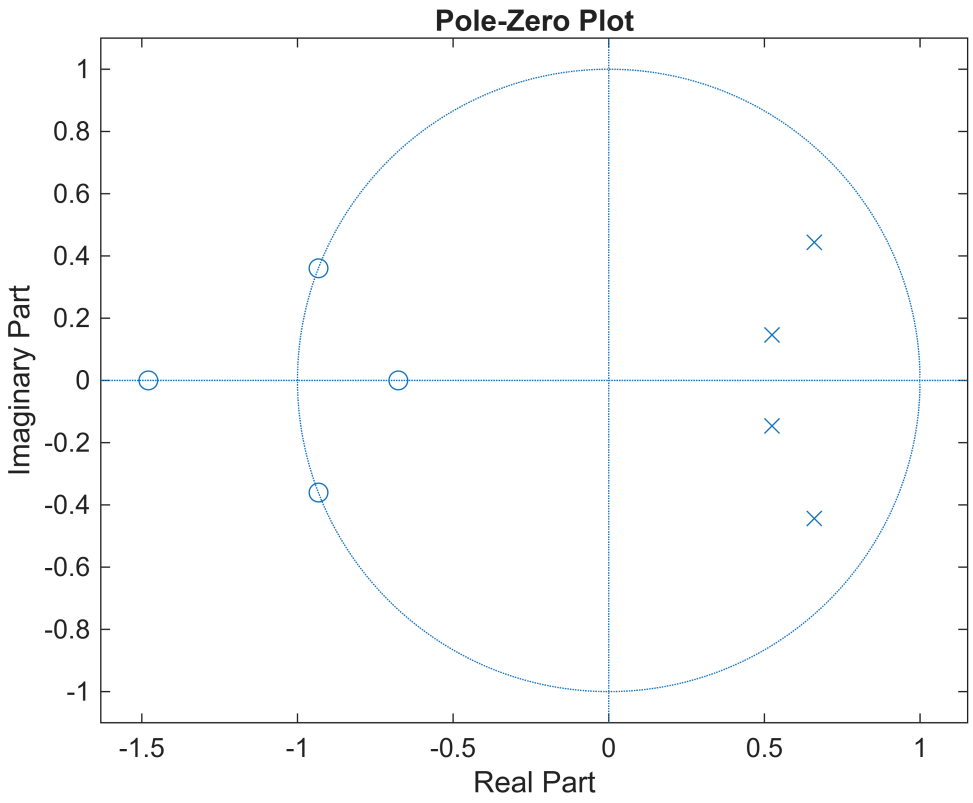

```
%% 7. Error Plot
figure;
plot(n,abs(y1-y2),'--','LineWidth',1.2); hold on;
plot(n,abs(y1-yc),':','LineWidth',1.5);
plot(n,abs(y1-yp),'-.','LineWidth',1.2);
grid on;
xlabel('n');
ylabel('Absolute Error');
title('Numerical Error');
legend('DF-I vs DF-II','DF-I vs Cascade','DF-I vs Parallel');
```




```
%% 8. Frequency Response
figure;
freqz(b,a,1024);
title('Frequency Response of IIR Filter');
```




```
%% 9. Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');
```


Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1